Problem · Unit 4 · 10 points · 60 min · Standard exam level

P07 — Insertion orbit and circularisation impulse: conic from the energy, apsides, apogee burn and time of flight

A probe P of mass m moves in the gravitational field of a planet of centre O, radius R and gravitational parameter μ: the only force is $\mathbf{F} = -\dfrac{m\mu}{r^2}\mathbf{e}_r$, with $r = |\mathbf{OP}|$, $\mathbf{e}_r = \mathbf{OP}/r$; O is fixed and the frame S_0 attached to it is inertial. At $t = 0$ the probe is at the distance r_0 from O and its velocity $\mathbf{v}_0$ has the magnitude

$$v_0 = \sqrt{\mu / r_0}$$

*(the local circular speed) and makes the **flight-path angle** $\gamma = 30^\circ$ with the local horizontal, pointing away from the planet. Use polar coordinates (r, θ) in the orbital plane, with θ (the true anomaly) measured from the perigee Π of the conic in the sense of the motion (Π is a virtual point: with the data below it lies inside the planet, so the probe never reaches it; the trajectory arc starts at the surface $r = R$), the unit vectors $\mathbf{e}_r$, $\mathbf{e}_\theta$ and $h = r^2\dot\theta > 0$. The local horizontal at P is the direction of $\mathbf{e}_\theta$, so $\mathbf{v}_0 = v_0(\sin\gamma\,\mathbf{e}_r + \cos\gamma\,\mathbf{e}_\theta)$.*

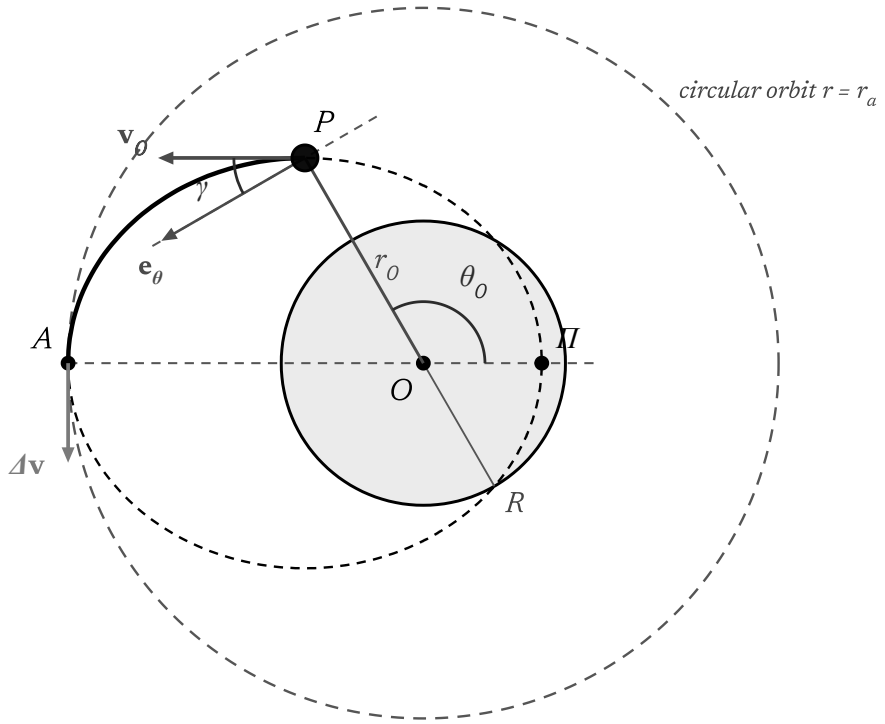


Figure 1. Insertion orbit: focus O (planet of radius R), probe P at r_0 with velocity $\mathbf{v}_0$ and flight-path angle γ above the local horizontal $\mathbf{e}_\theta$; true anomaly θ_0 from the perigee Π; apogee A, impulse $\Delta\mathbf{v}$ and the circular target orbit $r = r_a$ (dashed).

*When the probe reaches the apogee A, its engine fires a short **tangential** impulse $\Delta\mathbf{v}$ that puts it on the circular orbit of radius $r_a = |\mathbf{OA}|$.*

Setup

- Data: $\mu, r_0, v_0 = \sqrt{\mu/r_0}, \gamma = 30°$. Numerical part (f): $\mu = 4.0 \times 10^{14}\ \mathrm{m^3\ s^{-2}}, r_0 = 1.0 \times 10^7\ \mathrm{m}$, $R = 6.0 \times 10^6\ \mathrm{m}$.
- Hypotheses: point particle; planet's field spherically symmetric; no drag; the impulse is instantaneous (the position does not change during the burn) and the mass of propellant is irrelevant for the kinematics.
- Frames: $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{i}$ towards Π, $\mathbf{j}$ in the orbital plane and $\mathbf{k}$ along $\mathbf{H}_O$; polar basis $\mathbf{e}_r = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$, $\mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$. Change of basis:

$$\left(\mathbf{e}_r\ \mathbf{e}_\theta\ \mathbf{k}\right) = [R]\left(\mathbf{i}\ \mathbf{j}\ \mathbf{k}\right), \qquad [R] = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

with $[R][R]^T = I$ (rows of length $\cos^2\theta + \sin^2\theta = 1$, rows 1 and 2 orthogonal: $-\cos\theta\sin\theta + \sin\theta\cos\theta = 0$) and $\det[R] = \cos^2\theta + \sin^2\theta = +1$, so $\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}$.

- Results from Notes §8.4 used: orbit $r = p/(1 + e\cos\theta)$ with $p = h^2/\mu$ (Eq. 8.14), $p = a(1 - e^2)$ (Eq. 8.16), $r_p = a(1 - e)$, $r_a = a(1 + e)$, period $\tau = 2\pi\sqrt{a^3/\mu}$ (Eq. 8.18); vis-viva $v^2 = \mu(2/r - 1/a)$ (Notes §8.5).

Part (a) · 1.5 points

Conserved quantities and type of conic

Explain why h and the energy per unit mass E/m are constant. Compute them from the initial conditions and decide the type of conic and its semi-major axis a.

Step 1. Why they are conserved

$\mathbf{F}$ is central, so its torque about O is $\mathbf{r} \times \mathbf{F} = \mathbf{0}$ and $\mathbf{H}_O$ is constant (Notes §4.8); the motion is planar and $h = r^2\dot\theta$ is constant (Notes §4.8.1). $\mathbf{F} = -\nabla V$ with $V = -m\mu/r$, and it is the only force, so the mechanical energy $E = \frac{1}{2}mv^2 - m\mu/r$ is constant (Notes §4.5).

Step 2. Angular momentum

With $\mathbf{r} = r_0\,\mathbf{e}_r$ and $\mathbf{v}_0 = v_0(\sin\gamma\,\mathbf{e}_r + \cos\gamma\,\mathbf{e}_\theta)$:

$$\mathbf{r} \times \mathbf{v}_0 = r_0 v_0 \cos\gamma\,(\mathbf{e}_r \times \mathbf{e}_\theta) = r_0 v_0 \cos\gamma\,\mathbf{k} \implies h = r_0\sqrt{\frac{\mu}{r_0}}\,\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}\sqrt{\mu r_0}$$

Step 3. Energy and conic

$$\frac{E}{m} = \frac{1}{2}\,\frac{\mu}{r_0} - \frac{\mu}{r_0} = -\frac{\mu}{2r_0} < 0$$

A negative energy means a bounded orbit: an **ellipse**. The vis-viva equation (Notes §8.5) gives $E/m = -\mu/(2a)$, so $a = r_0$. Note the trap: the probe has exactly the circular speed of its radius, yet the orbit is not a circle, because the velocity is not horizontal; the speed fixes a only, the direction fixes the shape.

KEY RESULTS

- $h = r_0 v_0\cos\gamma = \frac{\sqrt{3}}{2}\sqrt{\mu r_0},\ h^2 = \frac{3}{4}\mu r_0$.
- $E/m = \frac{1}{2}v_0^2 - \mu/r_0 = -\dfrac{\mu}{2r_0} < 0$: ellipse, $a = -\dfrac{\mu}{2E/m} = r_0$.

Part (b) · 2 points

Shape of the orbit and apsides

Find the semi-latus rectum p, the eccentricity e and the perigee and apogee radii r_p, r_a. Verify the apsides independently from the energy equation.

Step 1. Semi-latus rectum

$$p = \frac{h^2}{\mu} = \frac{\frac{3}{4}\mu r_0}{\mu} = \frac{3}{4} r_0$$

Step 2. Eccentricity

From $p = a(1 - e^2)$ (Notes Eq. 8.16): $e^2 = 1 - p/a = 1 - \frac{3}{4} = \frac{1}{4}$, so $e = \frac{1}{2}$. Substituting $a = -\mu/(2E/m)$ and $p = h^2/\mu$ gives the general formula $e^2 = 1 + 2(E/m)h^2/\mu^2 = 1 + 2\big(-\frac{\mu}{2r_0}\big)\frac{3\mu r_0}{4\mu^2} = 1 - \frac{3}{4}$, the same value.

Step 3. Apsides

$$r_p = a(1 - e) = \frac{r_0}{2}, \qquad r_a = a(1 + e) = \frac{3r_0}{2}$$

and indeed $r_p + r_a = 2a$.

Step 4. Independent check

With $v^2 = \dot r^2 + h^2/r^2$, the energy at an apsis ($\dot r = 0$) reads $\dfrac{h^2}{2r^2} - \dfrac{\mu}{r} = \dfrac{E}{m}$. Multiplying by $2r^2/\mu$ with $h^2 = \frac{3}{4}\mu r_0$ and $E/m = -\mu/(2r_0)$:

$$\frac{3}{4} r_0 - 2r = -\frac{r^2}{r_0} \implies r^2 - 2r_0 r + \frac{3}{4} r_0^2 = 0 \implies r = r_0 \pm \sqrt{r_0^2 - \tfrac{3}{4} r_0^2} = r_0 \pm \frac{r_0}{2}$$

the same $r_p = r_0/2$ and $r_a = 3r_0/2$.

KEY RESULTS

- $p = \frac{3}{4} r_0,\ e = \sqrt{1 - p/a} = \frac{1}{2}$ (equivalently $e^2 = 1 + 2(E/m)h^2/\mu^2$).
- $r_p = a(1 - e) = \frac{1}{2} r_0,\ r_a = a(1 + e) = \frac{3}{2} r_0$.

Part (c) · 1.5 points

Position on the orbit, apsidal speeds and period

Find the true anomaly θ_0 of the initial point (justify the sign), the speeds v_p, v_a at the apsides of the conic (v_p is the speed the conic would have at its virtual perigee) and the period τ of the full ellipse.

Step 1. True anomaly

$$r_0 = \frac{p}{1+e\cos\theta_0} \implies 1+\tfrac{1}{2}\cos\theta_0 = \frac{p}{r_0} = \frac{3}{4} \implies \cos\theta_0 = -\frac{1}{2}$$

so $\theta_0 = \pm 120^\circ$.

Step 2. Sign from the radial velocity

Differentiating $r = p\,(1+e\cos\theta)^{-1}$ with the chain rule and $\dot\theta = h/r^2$:

$$\dot r = \frac{p\,e\sin\theta\,\dot\theta}{(1+e\cos\theta)^2} = \frac{r^2}{p}\,e\sin\theta\,\frac{h}{r^2} = \frac{h}{p}e\sin\theta = \frac{\mu}{h}e\sin\theta$$

The velocity points away from the planet, $\dot r = v_0\sin\gamma > 0$, so $\sin\theta_0 > 0$ and $\theta_0 = +120^\circ$. Check: $\frac{\mu}{h}\,\frac{1}{2}\,\frac{\sqrt3}{2} = \frac{\mu}{\sqrt3\sqrt{\mu r_0}}\,\frac{\sqrt3}{2} = \frac{1}{2}\sqrt{\mu/r_0} = v_0\sin 30^\circ$.

Step 3. Apsidal speeds

At the apsides the velocity is perpendicular to $\mathbf{r}$ ($\dot r = 0$), so $v = h/r$:

$$v_p = \frac{h}{r_p} = \frac{\frac{\sqrt3}{2}\sqrt{\mu r_0}}{r_0/2} = \sqrt3\sqrt{\frac{\mu}{r_0}}, \qquad v_a = \frac{h}{r_a} = \frac{\frac{\sqrt3}{2}\sqrt{\mu r_0}}{3r_0/2} = \frac{1}{\sqrt3}\sqrt{\frac{\mu}{r_0}}$$

Vis-viva check at the apogee: $\mu\big(\frac{2}{r_a}-\frac{1}{a}\big) = \frac{\mu}{r_0}\big(\frac{4}{3}-1\big) = \frac{\mu}{3r_0} = v_a^2$.

Step 4. Period

Kepler's third law (Notes Eq. 8.18) with $a = r_0$: $\tau = 2\pi\sqrt{a^3/\mu} = 2\pi\sqrt{r_0^3/\mu}$.

KEY RESULTS

- $\cos\theta_0 = -\frac{1}{2}$ and $\dot r > 0$, so $\theta_0 = +120^\circ$ (outbound, between perigee and apogee).
- $v_p = h/r_p = \sqrt3\sqrt{\mu/r_0}$, $v_a = h/r_a = \sqrt{\mu/r_0}/\sqrt3$; $\tau = 2\pi\sqrt{r_0^3/\mu}$.

Part (d) · 2 points

Circularisation impulse at the apogee

Compute the impulse $\Delta\mathbf{v}$ (magnitude and direction) that makes the orbit circular at r_a, the change of energy per unit mass and the ratio of the new period to τ. Why is the apogee the right place for a purely tangential burn?

Step 1. Why the apogee

A circular orbit has $\dot r = 0$ everywhere, so its velocity is perpendicular to $\mathbf{r}$. At the apogee $\dot r = 0$ already: the velocity $v_a\,\mathbf{e}_\theta$ is horizontal and only its magnitude has to change. At any other point the burn would also have to cancel $\dot r$, so it could not be purely tangential.

Step 2. Impulse

The circular speed at r_a follows from $\mu/r_a^2 = v^2/r_a$ (or from vis-viva with $a = r_a$):

$$v_{circ} = \sqrt{\frac{\mu}{r_a}} = \sqrt{\frac{2\mu}{3r_0}}, \qquad \Delta\mathbf{v} = (v_{circ} - v_a)\,\mathbf{e}_\theta = \left(\sqrt{\tfrac{2}{3}} - \tfrac{1}{\sqrt{3}}\right)\sqrt{\frac{\mu}{r_0}}\,\mathbf{e}_\theta = \frac{\sqrt{2}-1}{\sqrt{3}}\sqrt{\frac{\mu}{r_0}}\,\mathbf{e}_\theta$$

positive: the engine pushes **forward**, in the direction of motion.

Step 3. Energy

The position does not change during the burn, so only the kinetic energy changes:

$$\frac{\Delta E}{m} = \frac{1}{2}\left(v_{circ}^2 - v_a^2\right) = \frac{1}{2}\left(\frac{2}{3} - \frac{1}{3}\right)\frac{\mu}{r_0} = \frac{\mu}{6r_0}$$

Check with $E/m = -\mu/(2a)$: $-\dfrac{\mu}{2r_a} + \dfrac{\mu}{2a} = -\dfrac{\mu}{3r_0} + \dfrac{\mu}{2r_0} = \dfrac{\mu}{6r_0}$.

Step 4. New period

$\tau_{circ} = 2\pi\sqrt{r_a^3/\mu}$, so $\tau_{circ}/\tau = (r_a/a)^{3/2} = (3/2)^{3/2} \approx 1.837$.

KEY RESULTS

- $v_{circ} = \sqrt{\mu/r_a} = \sqrt{2/3}\,\sqrt{\mu/r_0}$; $\Delta v = v_{circ} - v_a = \dfrac{\sqrt{2}-1}{\sqrt{3}}\sqrt{\dfrac{\mu}{r_0}} \approx 0.239\sqrt{\mu/r_0}$, along $+\mathbf{e}_\theta$ (prograde).
- $\Delta E/m = \tfrac{1}{2}(v_{circ}^2 - v_a^2) = \dfrac{\mu}{6r_0} > 0$; $\tau_{circ}/\tau = (3/2)^{3/2} \approx 1.837$.

Part (e) · 1.5 points

Time of flight from r_0 to the apogee

*Compute the time the probe needs to go from the initial point to the apogee. Show that it is **not** $\tau \cdot (60^\circ/360^\circ)$. (Extension: use the eccentric anomaly $\mathcal{E}$, defined by $r = a(1 - e\cos\mathcal{E})$.)*

Step 1. Eccentric anomaly (extension)

Define $\mathcal{E}$ by $r = a(1 - e\cos\mathcal{E})$; it runs from 0 at the perigee to π at the apogee. Equating with $r = p/(1 + e\cos\theta)$ and $p = a(1 - e^2)$ and solving for $\cos\mathcal{E}$:

$$\cos\mathcal{E} = \frac{1}{e}\Big(1 - \frac{1 - e^2}{1 + e\cos\theta}\Big) = \frac{e + \cos\theta}{1 + e\cos\theta}$$

At $\theta_0 = 120^\circ$: $\cos\mathcal{E}_0 = (\tfrac{1}{2} - \tfrac{1}{2})/(1 - \tfrac{1}{4}) = 0$, so $\mathcal{E}_0 = \pi/2$ (outbound branch); check $r = a(1 - 0) = r_0$. At the apogee $\mathcal{E} = \pi$.

Step 2. Kepler's equation

The energy equation $\dot r^2 + h^2/r^2 = \mu(2/r - 1/a)$ with $h^2 = \mu a(1 - e^2)$ gives

$$\dot r^2 = \frac{\mu}{a\,r^2}\big[2ar - r^2 - a^2(1 - e^2)\big]$$

Substituting $r = a(1 - e\cos\mathcal{E})$: $2ar - r^2 - a^2(1 - e^2) = a^2\big[2 - 2e\cos\mathcal{E} - 1 + 2e\cos\mathcal{E} - e^2\cos^2\mathcal{E} - 1 + e^2\big] = a^2e^2\sin^2\mathcal{E}$. Hence $\dot r = \pm\sqrt{\mu/a}\;a\,e\sin\mathcal{E}/r$. With $\mathcal{E}$ increasing along the motion, $\sin\mathcal{E} > 0$ on the outbound branch ($0 < \mathcal{E} < \pi, \dot r > 0$) and $\sin\mathcal{E} < 0$ on the inbound branch ($\pi < \mathcal{E} < 2\pi, \dot r < 0$), so the sign is $+$ on **both** branches: $\dot r = \sqrt{\mu/a}\;a\,e\sin\mathcal{E}/r$ for every $\mathcal{E}$. On the other hand, by the chain rule $\dot r = a\,e\sin\mathcal{E}\;\dot{\mathcal{E}}$, also valid on both branches. Equating (and, at the apses where $\sin\mathcal{E} = 0$, by continuity):

$$\dot{\mathcal{E}} = \frac{\sqrt{\mu/a}}{r} = \frac{n}{1 - e\cos\mathcal{E}} \quad \text{for every } \mathcal{E} \implies (1 - e\cos\mathcal{E})\,d\mathcal{E} = n\,dt, \qquad n = \sqrt{\frac{\mu}{a^3}}$$

Step 3. Integration with Barrow's rule

The antiderivative of $1 - e\cos\mathcal{E}$ is $\mathcal{E} - e\sin\mathcal{E}$, so

$$n\,t_{0\to A} = \big[\mathcal{E} - e\sin\mathcal{E}\big]_{\pi/2}^{\pi} = (\pi - 0) - \Big(\frac{\pi}{2} - \frac{1}{2}\Big) = \frac{\pi}{2} + \frac{1}{2}$$

and with $1/n = \sqrt{r_0^3/\mu}$ (because $a = r_0$): $t_{0\to A} = \left(\frac{\pi}{2} + \frac{1}{2}\right)\sqrt{r_0^3/\mu} \approx 2.071\sqrt{r_0^3/\mu}$. As a fraction of the period, $(\pi/2 + 1/2)/(2\pi) \approx 0.330$.

Step 4. The trap

The angle swept is $60°$, one sixth of a turn, but the time is a third of the period: near the apogee the probe is slow ($\dot\theta = h/r^2$ is small), so equal angles do **not** take equal times. A direct numerical quadrature of $dt = r^2\, d\theta/h$ from $120°$ to $180°$ gives the same $\pi/2 + 1/2$.

KEY RESULTS

- $\cos\mathcal{E} = \dfrac{e + \cos\theta}{1 + e\cos\theta}$: $\mathcal{E}_0 = 90°$ at $\theta_0 = 120°$, $\mathcal{E} = 180°$ at the apogee.
- $t_{0\to A} = \dfrac{1}{n}\left[\mathcal{E} - e\sin\mathcal{E}\right]_{\pi/2}^{\pi} = \left(\dfrac{\pi}{2} + \dfrac{1}{2}\right)\sqrt{\dfrac{r_0^3}{\mu}} \approx 0.330\,\tau$, twice the naive $\tau/6 \approx 0.167\,\tau$.

Part (f) · 1.5 points

Numerical values and a failed burn

With $\mu = 4.0 \times 10^{14}\ m^3\ s^{-2}$, $r_0 = 1.0 \times 10^7\ m$: compute v_0, h, v_p, v_a, τ, Δv, $\Delta E/m$ and $t_{0\to A}$. If the engine had failed at the apogee, would the probe have survived, given the planet radius $R = 6.0 \times 10^6\ m$? If not, at which true anomaly and how long after the apogee would it hit the surface?

Step 1. Numbers

$\sqrt{\mu/r_0} = \sqrt{4.0 \times 10^7} = 6325$ m/s $= v_0$ and $\sqrt{r_0^3/\mu} = \sqrt{2.5 \times 10^6} = 1581$ s. Then $h = \frac{\sqrt{3}}{2} r_0 v_0 = 5.477 \times 10^{10}$ m^2/s, $v_p = \sqrt{3} \times 6325 = 1.095 \times 10^4$ m/s, $v_a = 6325/\sqrt{3} = 3651$ m/s, $\tau = 2\pi \times 1581 = 9935$ s, $v_{circ} = 5164$ m/s, $\Delta v = 5164 - 3651 = 1512$ m/s, $\Delta E/m = \mu/(6r_0) = 6.67 \times 10^6$ J/kg, $\tau_{circ} = 1.825 \times 10^4$ s and $t_{0\to A} = 2.071 \times 1581 = 3274$ s, against the naive $\tau/6 = 1656$ s.

Step 2. Failed burn

$r_p = r_0/2 = 5.0 \times 10^6$ m is smaller than $R = 6.0 \times 10^6$ m: after the apogee the probe would descend into the planet. Impact: $7.5 \times 10^6/(1 + \frac{1}{2}\cos\theta) = 6.0 \times 10^6 \Rightarrow \cos\theta = \frac{1}{2}$, and on the inbound branch ($\dot r < 0$, $\sin\theta < 0$) $\theta = 300°$. There $\cos\mathcal{E} = (\frac{1}{2} + \frac{1}{2})/(1 + \frac{1}{4}) = \frac{4}{5}$ with $\mathcal{E}$ in $(\pi, 2\pi)$: $\mathcal{E}_i = 2\pi - \arccos\frac{4}{5} = 5.640$ rad, $\sin\mathcal{E}_i = -\frac{3}{5}$.

Step 3. Time to impact

$$n\,t_{A\to i} = \left[\mathcal{E} - e\sin\mathcal{E}\right]_{\pi}^{\mathcal{E}_i} = \left(5.640 + \frac{1}{2}\cdot\frac{3}{5}\right) - \pi = 2.798 \implies t_{A\to i} = 2.798 \times 1581 = 4424\ \text{s}$$

The circularisation burn is therefore not optional: it also raises the perigee from $0.5\,r_0$ to $1.5\,r_0$.

KEY RESULTS

- $v_0 = 6325$ m/s, $h = 5.477 \times 10^{10}$ m^2/s, $v_p = 1.095 \times 10^4$ m/s, $v_a = 3651$ m/s, $\tau = 9935$ s, $\Delta v = 1512$ m/s, $\Delta E/m = 6.67 \times 10^6$ J/kg, $t_{0\to A} = 3274$ s (naive: 1656 s).
- $r_p = 5.0 \times 10^6$ m $< R$: impact at $\theta = 300^\circ$ ($\cos\theta = \frac{1}{2}$, $\cos\mathcal{E} = \frac{4}{5}$), 4424 s after the apogee.

Interpretation and checks

- **Dimensional check:** $h = r_0 v_0$ in m^2/s; h^2/μ in m^4 s^{-2}/(m^3 s^{-2}) = m; E/m and μ/r in m^2 s^{-2} = J/kg; $\sqrt{a^3/\mu}$ in s; Δv in m/s.
- **Limits:** $\gamma \to 0$ with $v_0 = \sqrt{\mu/r_0}$ gives $h^2 = \mu r_0$, $e = 0$: the probe is already on the circle of radius r_0. $\gamma \to 90^\circ$ gives $h = 0$: a radial (degenerate) orbit, straight up and back down. For any γ, $a = r_0$, so the period does not depend on γ.
- **Common mistakes in the exam:** (1) concluding that the orbit is circular because $v_0 = \sqrt{\mu/r_0}$; (2) using $h = r_0 v_0$ without $\cos\gamma$; (3) taking $\theta_0 = -120^\circ$ (the inbound point) without checking the sign of $\dot r$; (4) computing Δv as $v_{circ}(r_0) - v_0$ or applying it at r_0; (5) converting angles into times with $\tau\,\Delta\theta/2\pi$, which is wrong for any $e \neq 0$.